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Subject: Programming with Java

Sem: III AY: 2026-27

Assignment No: 10

TITLE: Write a Java program to demonstrate the use of abstract classes and abstract methods

Problem Statement

Write a Java program to demonstrate the use of abstract classes and abstract methods.

Learning Objectives

To implement abstraction using abstract classes and abstract methods.

Theory

Abstract classes provide partial implementation of a program and cannot be instantiated directly. They may contain both abstract and concrete methods. Abstract methods declare functionality that must be implemented by subclasses. Abstraction hides implementation details while exposing only essential functionality. This improves code flexibility and maintainability in object-oriented programming.

Exercise

1. Create an abstract **Payment class** and implement Credit Card and UPI payment classes.?

```
1  abstract class Payment
2  {
3      abstract void pay();
4
5      void display()
6      {
7          System.out.println("This is a Payment");
8      }
9  }
10
11  // Child class
12  class CreditCard extends Payment
13  {
14      void pay()
15      {
16          System.out.println("Payment made using Credit Card");
17      }
18  }
19
20  // Child class
21  class UPI extends Payment
22  {
23      void pay()
24      {
25          System.out.println("Payment made using UPI");
26      }
27  }
28
29  public class PaymentDemo
30  {
31      Run | Debug
32      public static void main(String[] args)
33      {
34          CreditCard c = new CreditCard();
35          c.display();
36          c.pay();
37
38          UPI u = new UPI();
39          u.display();
40          u.pay();
41      }
42  }
```

Output

```
[Running] cd "/Users/sumit/Documents/java/java1/" && javac PaymentDemo.java && java PaymentDemo
This is a Payment
Payment made using Credit Card
This is a Payment
Payment made using UPI

[Done] exited with code=0 in 0.285 seconds
```

2. Create an abstract **FoodOrder** class with an abstract method `calculateBill()`. Implement two subclasses, **DineInOrder** and **TakeAwayOrder**, to calculate and display the total bill based on the order type.

```
Users > sumit > Documents > java > java1 > J FoodOrderDemo.java > ⚙ TakeAwayOrder
1  abstract class FoodOrder
2  {
3      abstract void calculateBill();
4
5      void display()
6      {
7          System.out.println("This is a Food Order");
8      }
9  }
10
11 // Child class
12 class DineInOrder extends FoodOrder
13 {
14     void calculateBill()
15     {
16         double foodPrice = 500;
17         double serviceCharge = 50;
18
19         System.out.println("Dine-In Order Bill : " + (foodPrice + serviceCharge));
20     }
21 }
22 💡
23 // Child class
24 class TakeAwayOrder extends FoodOrder
25 {
26     void calculateBill()
27     {
28         double foodPrice = 500;
29         double packingCharge = 30;
30
31         System.out.println("Take-Away Order Bill : " + (foodPrice + packingCharge));
32     }
33 }
34
```

```

35 public class FoodOrderDemo
36 {
    Run | Debug
37     public static void main(String[] args)
38     {
39         DineInOrder d = new DineInOrder();
40         d.display();
41         d.calculateBill();
42
43         TakeAwayOrder t = new TakeAwayOrder();
44         t.display();
45         t.calculateBill();
46     }
47 }

```

Output:

```

[Running] cd "/Users/sumit/Documents/java/java1/" && javac FoodOrderDemo.java && java FoodOrderDemo
This is a Food Order
Dine-In Order Bill : 550.0
This is a Food Order
Take-Away Order Bill : 530.0

[Done] exited with code=0 in 0.403 seconds

```

Conclusion:

The program successfully demonstrated the use of abstract classes and abstract methods in Java. An abstract class provides a common structure for its subclasses, while abstract methods must be implemented by the child classes. The programs showed how different subclasses can provide their own implementation of the same abstract method. Thus, abstraction helps achieve code reusability, flexibility, and better program organization.